

Full GPIO reference — 30-pin DevKit V1

GPIO	Class	Safe for	Your use
0	strapping, BOOT btn	input w/ pull-up; must be HIGH at boot	—
1	UART0 TX, boot-glitch	free only if you drop serial	—
2	strapping, onboard LED	output idle-LOW; LED/buzzer	—
3	UART0 RX	free only if you drop serial	—
4	clean I/O	anything	OLED DC
5	strapping (HIGH at boot) + boot-glitch	idle-HIGH only: relay, CLK, I ² C. Not IR TX	spare
6–11	flash	never	—
12	strapping (LOW at boot)	idle-LOW only; best left NC	NC
13	clean I/O	anything	Light L1
14	boot-glitch	outputs where a boot pulse is harmless	TM1637 DIO
15	strapping (HIGH) + boot-glitch	idle-HIGH, glitch-tolerant	TM1637 CLK
16	clean I/O (WROOM only)	anything	OLED MOSI
17	clean I/O (WROOM only)	anything	IR TX
18	clean I/O	anything	OLED SCK
19	clean I/O	anything	Light L3
21	clean I/O	anything	Light L2
22	clean I/O	anything	I ² C SDA
23	clean I/O	anything	I ² C SCL
25	clean I/O + DAC1	anything, analog out	Fan F3
26	clean I/O + DAC2	anything, analog out	Fan F4
27	clean I/O	anything	Fan F5
32	clean I/O + ADC1	anything, analog in	Fan F1
33	clean I/O + ADC1	anything, analog in	Fan F2
34	input-only , ADC1	input + ext pull-up	IR RX
35	input-only , ADC1	input + ext pull-up	Fan +
36	input-only , ADC1	input + ext pull-up	Fan –
39	input-only , ADC1	input + ext pull-up; analog	spare

The classes in one line: **clean I/O** = 4, 13, 16–19, 21–23, 25–27, 32, 33 (all yours). **Glitch/strapping** = 2, 5, 14, 15. **Input-only** = 34–36, 39. **Never** = 6–11, and 12 in practice.